

Unit 1.10

Note Title

2012/02/27

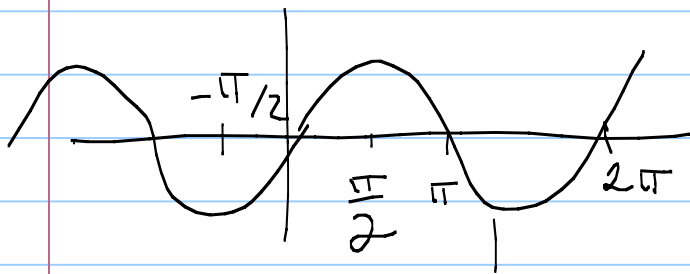
Inverse trigonometric function

* Is $\sin x$ or $\cos x$ one-to-one?

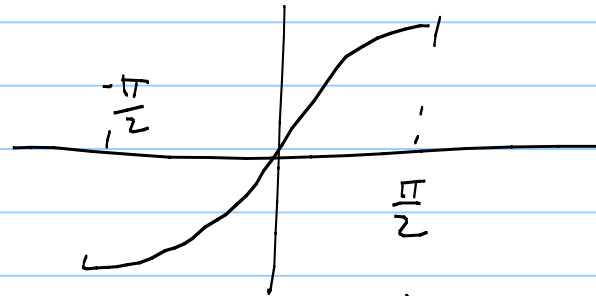
* Is $\sin x$, $-\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$

one-to-one?

Inverse Trigonometric function



$\sin x$



$\sin x,$

$x \in [-\pi/2, \pi/2]$

$$f^{-1}(x) = y \iff f(y) = x$$

The function $\arcsin$

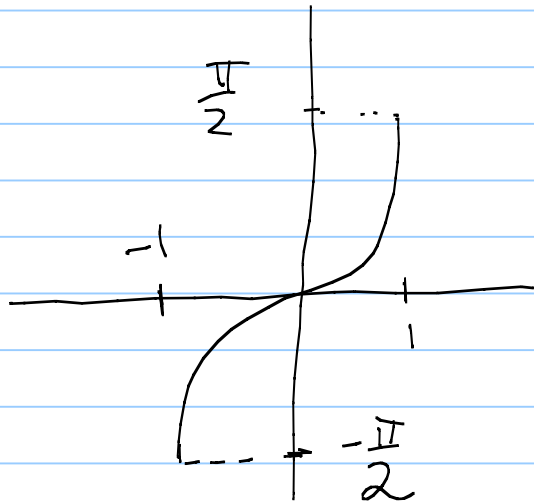
$$y = \sin^{-1} x = \arcsin x$$

i f f $\Leftrightarrow$

$$\sin y = x \quad \text{AND}$$

$$-\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$$

Sketch $y = \arcsin x$. What is the domain and range of $\arcsin x$



$\arcsin x$

Domain $x \in [-1, 1]$

Range $y \in [-\pi/2, \pi/2]$

Cancellation equation

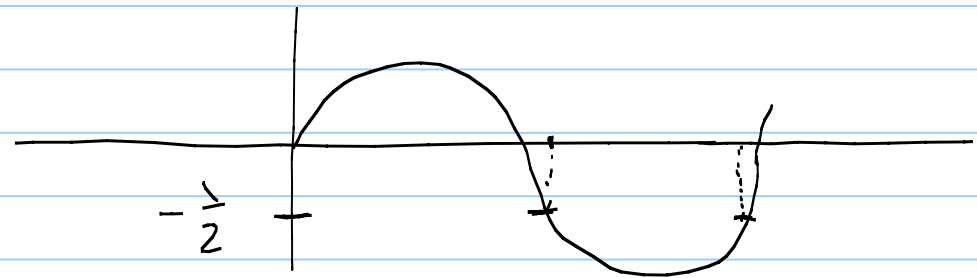
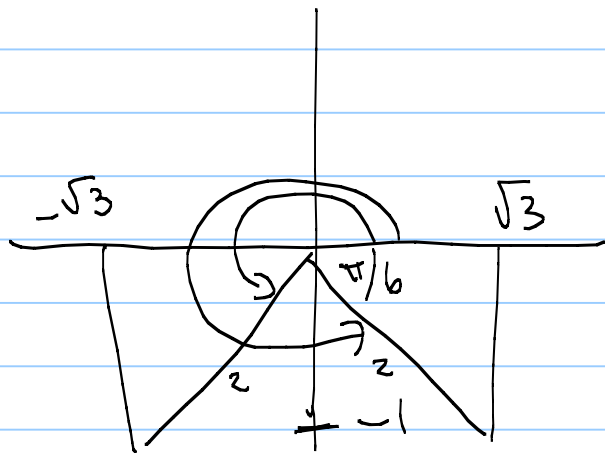
$$* \sin^{-1}(\sin x) = \arcsin(\sin x) = x$$

$$* \sin(\sin^{-1} x) = \sin(\arcsin x) = x$$

for $-\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$
for $-1 \leq x \leq 1$

Example

1) Solve $\sin x = -\frac{1}{2}$ (y/r)



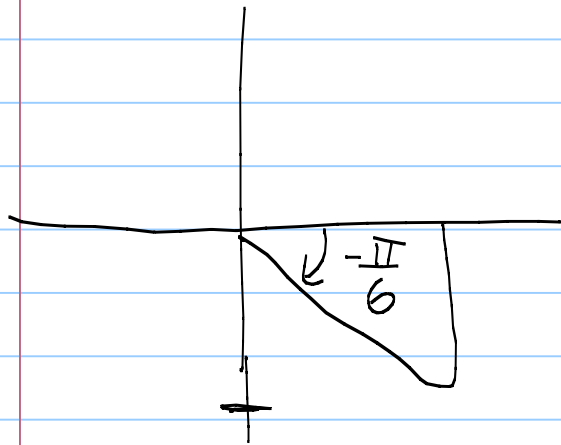
$$x = \pi + \pi/6 + 2k\pi, \quad k \in \mathbb{Z}$$

OR

$$\begin{aligned} x &= 2\pi - \pi/6 + 2k\pi, \quad k \in \mathbb{Z} \\ &= \frac{11\pi}{6} + 2k\pi, \quad k \in \mathbb{Z} \end{aligned}$$

2) Find $\arcsin(-\frac{1}{2})$

$$y = \arcsin(-\frac{1}{2}) \Rightarrow \sin y = -\frac{1}{2} \quad \text{AND} \\ y \in [-\pi/2, \pi/2]$$

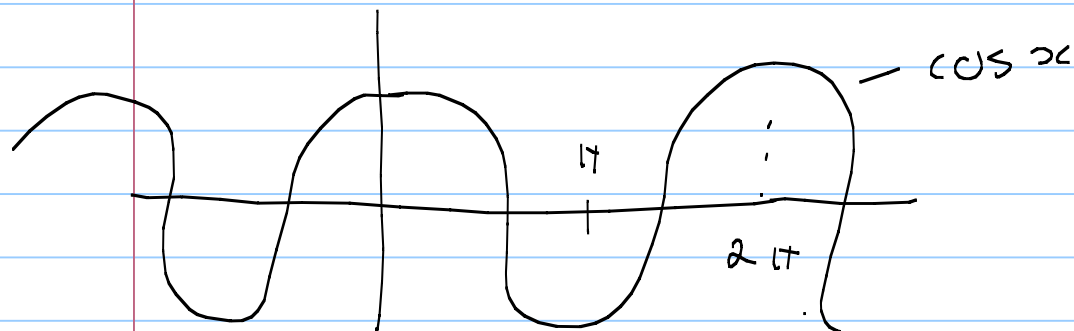


$$y = -\pi/6$$

3] Find
 $\arcsin(\sin(\pi/4))$

$$= \pi/4$$

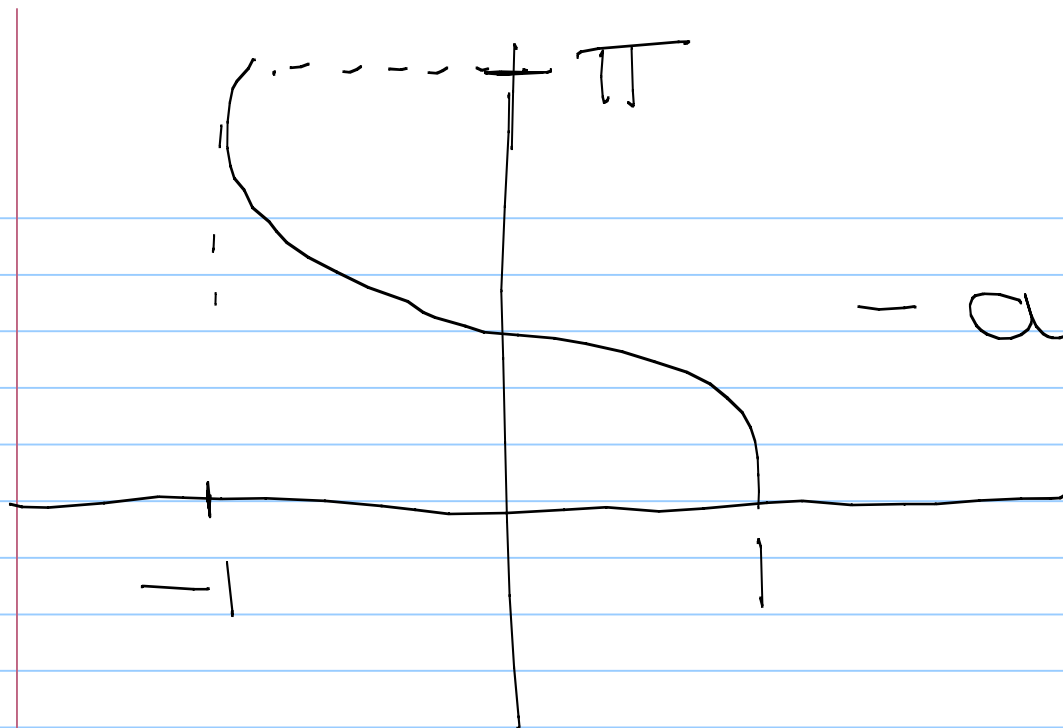
The function $\arccos$



restricting $\cos x$ on
interval $0 \leq x \leq \pi$.



← one-to-one.



$$- \arccos x = \cos^{-1} x$$

Domain : $x \in [-1, 1]$

Range : $y \in [0, \pi]$

$$\cos^{-1} x = y \Leftrightarrow \cos y = x \text{ AND } 0 \leq y \leq \pi$$

The two cancellation properties

$$1] \quad \cos^{-1} [\cos x] = x \quad \text{for } 0 \leq x \leq \pi$$

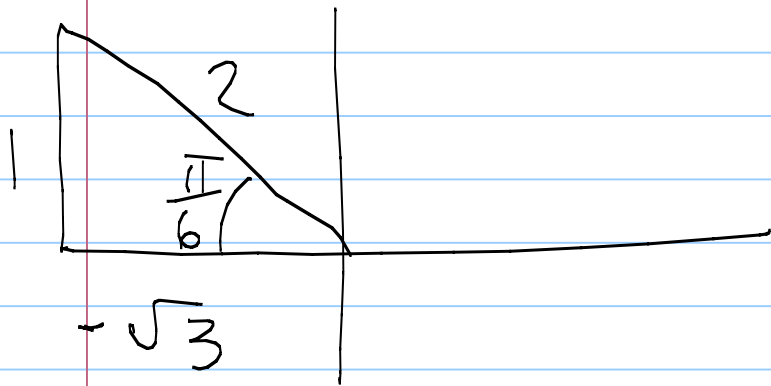
$$2] \quad \cos [\cos^{-1} x] = x \quad \text{for } -1 \leq x \leq 1$$

Example

1] Find $\arccos\left(-\frac{\sqrt{3}}{2}\right)$

$$y = \arccos\left(-\frac{\sqrt{3}}{2}\right) \Leftrightarrow \cos y = -\frac{\sqrt{3}}{2}$$

$$0 \leq y \leq \pi$$



$$y = \pi - \frac{\pi}{6} = \frac{5\pi}{6}$$

2] Find $\arccos(\cos(\frac{-\pi}{2}))$

$$\arccos(\cos x) = x, \quad 0 \leq x \leq \pi$$

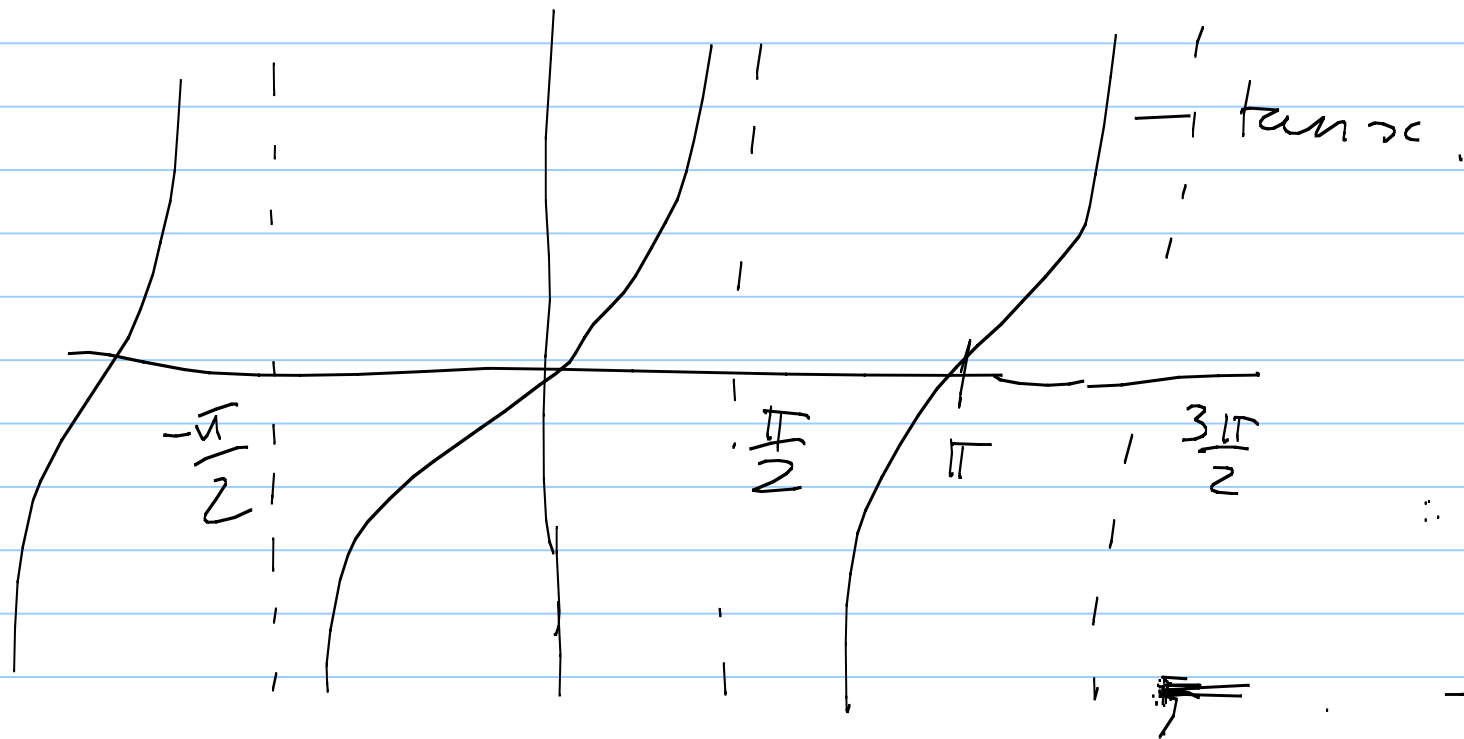
$$y = \arccos(\cos(-\pi/2))$$
$$= \arccos(0)$$

$$\Leftrightarrow \cos y = 0, \quad y \in [0, \pi]$$

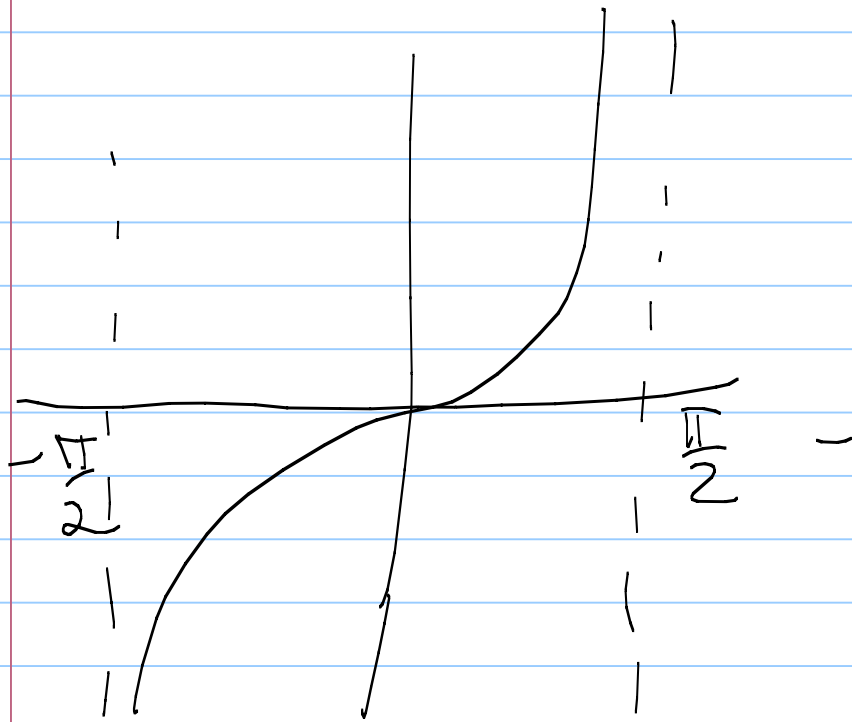
$$\Rightarrow y = \pi/2$$

$$\Rightarrow \arccos(\cos(-\pi/2)) = \pi/2$$

The function $\arctan$

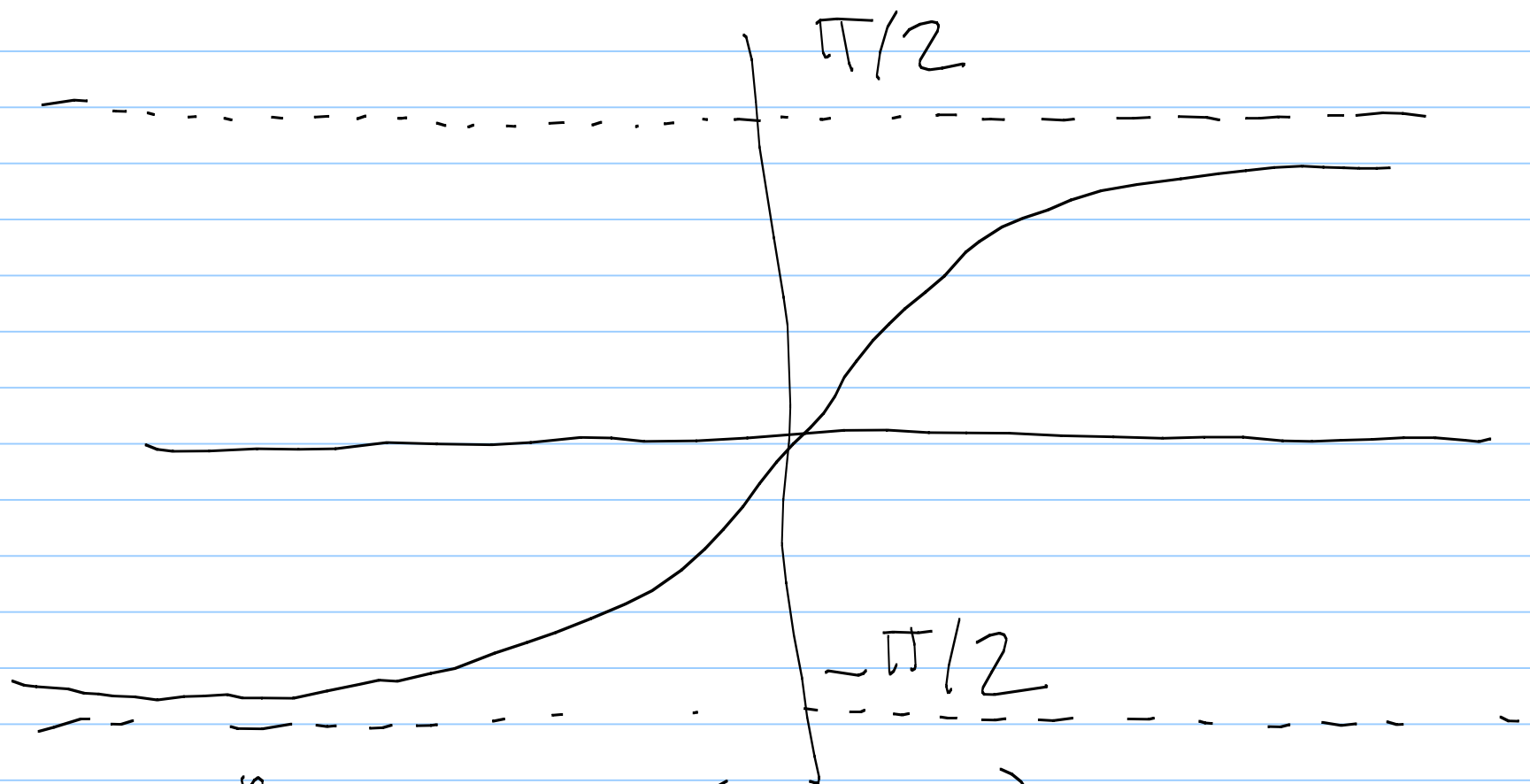


We work with $f(x) = \tan x$,
 $x \in (-\pi/2, \pi/2)$



Range
 $y \in (-\infty, \infty)$

Sketch $\arctan x = f(x)$



Domain: $x \in (-\infty, \infty)$

Range: $y \in (-\pi/2, \pi/2)$

The definition

$$y = \arctan x \iff \tan y = x$$

$$\text{AND } y \in (-\pi/2, \pi/2)$$

Cancellation Properties

$$\tan(\arctan(x)) = x \quad \text{if } x \in \mathbb{R}$$

$$\arctan(\tan(x)) = x \quad \text{if } x \in (-\frac{\pi}{2}, \frac{\pi}{2})$$

Example

1) Solve

$$\tan x = -1$$

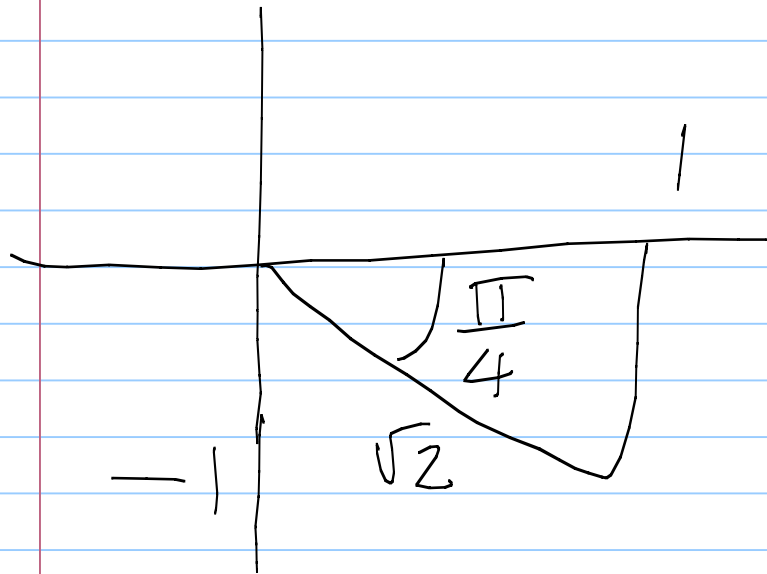
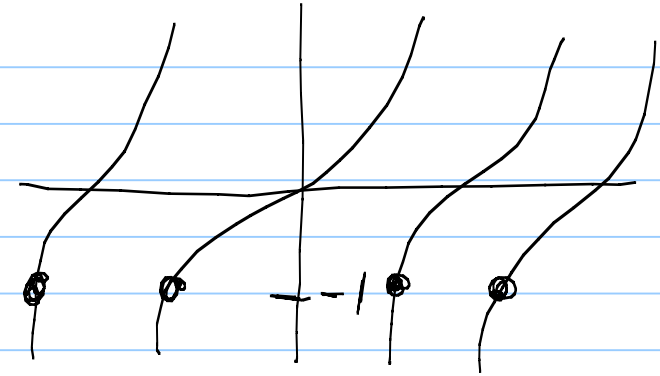
2) Find $\arctan(-1)$

3) Find $\cos(\arctan(-\frac{1}{5}))$

4) Simplify $\cos(\arctan x)$

Solutions

1) $\tan x = -1$ (y/x)



$$x = -\frac{\pi}{4} + k\pi, \quad k \in \mathbb{Z}$$

$$2) y = \arctan(-1)$$

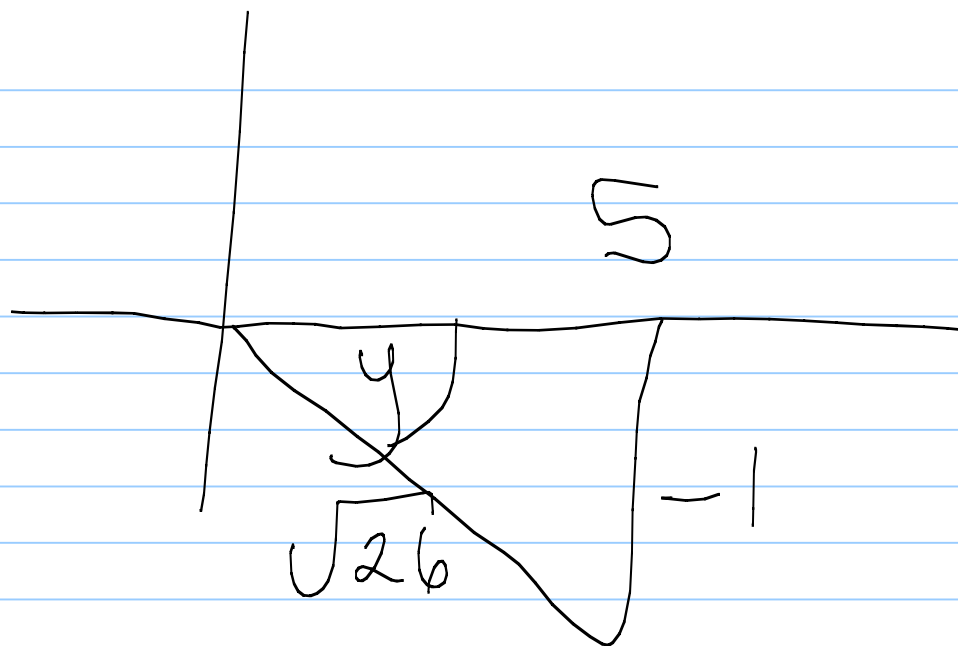
$$\Leftrightarrow \tan y = -1 \text{ and } -\frac{\pi}{2} < y < \frac{\pi}{2}$$

$$y = -\pi/4 \rightarrow$$

$$3) \cos(\arctan -\frac{1}{5})$$

$$\text{Let } y = \arctan(-\frac{1}{5})$$

$$\Leftrightarrow \tan y = -\frac{1}{5} \quad \text{and} \quad y \in (-\frac{\pi}{2}, \frac{\pi}{2})$$

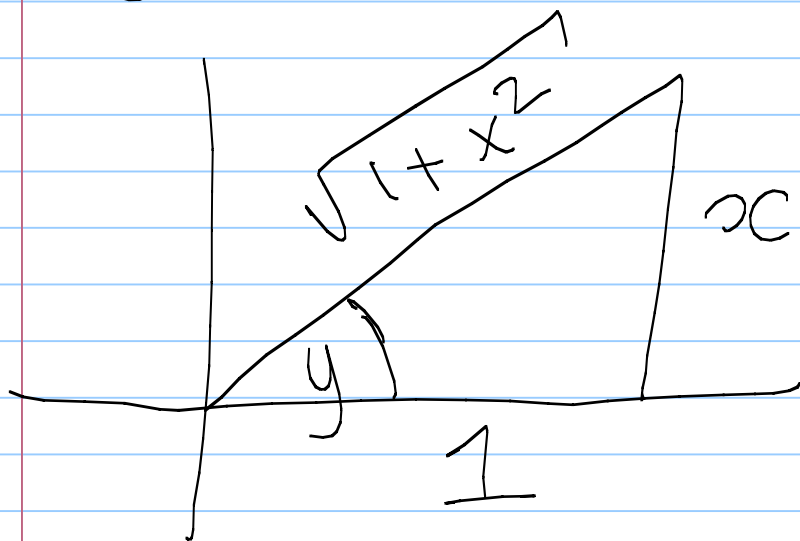


$$r = \sqrt{5^2 + (-1)^2}$$
$$= \sqrt{26}$$

$$\Rightarrow \cos(\arctan(-\frac{1}{5}))$$
$$= \cos y = \frac{5}{\sqrt{26}}$$

4) $\cos(\arctan x)$

$y = \arctan x \Leftrightarrow \tan y = x$ and
 $y \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$



Then
$$\cos y = \frac{1}{\sqrt{1+x^2}}$$

Example ✓

① $\sin(\arcsin \frac{1}{2})$

② $\cos^{-1}(\cos(\frac{4\pi}{4}))$

③ $\cos(\arcsin x)$

④ $\tan(\arcsin(\frac{1}{3}))$

$$\textcircled{1} \sin(\arcsin \frac{1}{2}) = \frac{1}{2}$$

$$\textcircled{2} \arccos(\cos \frac{11\pi}{4})$$

$$= \arccos(\cos(\frac{11\pi}{4} - \frac{8\pi}{4}))$$

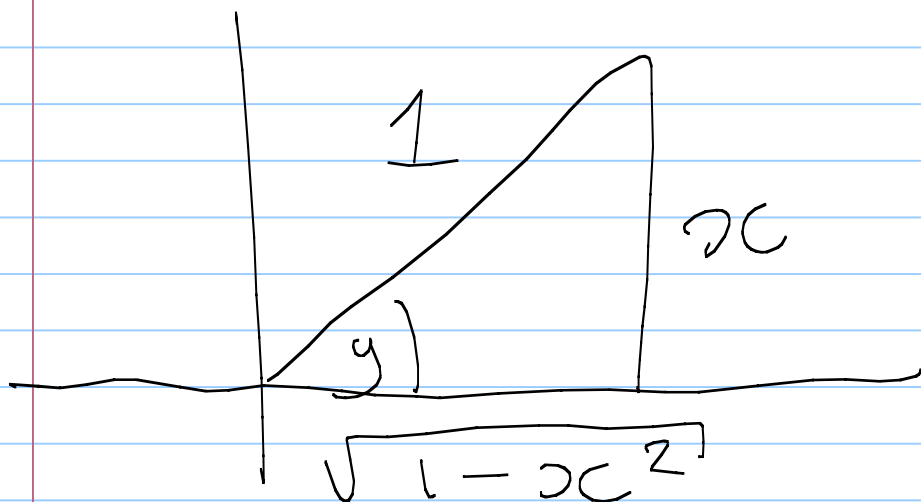
$$= \arccos(\cos(3\pi/4))$$

$$= 3\pi/4$$

$$\textcircled{3} \cos(\arcsin x)$$

$$\text{Let } y = \arcsin x$$

$$\Leftrightarrow \sin y = x \text{ and } y \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$$



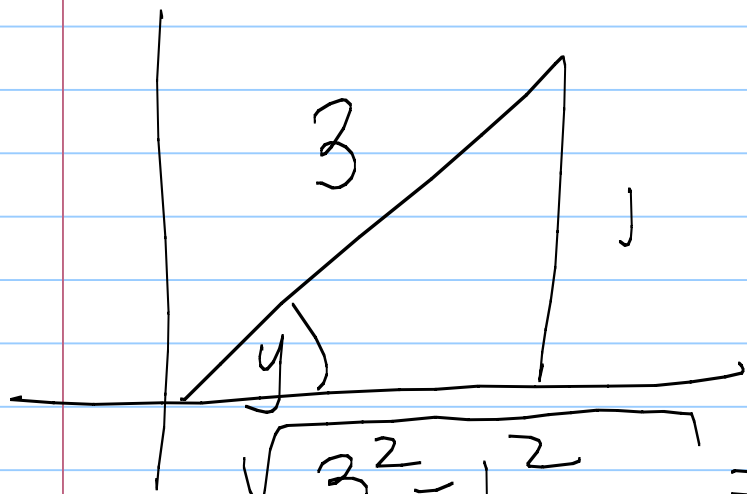
Then

$$\cos y = \frac{\sqrt{1-x^2}}{1}$$

$$\textcircled{4} \quad \tan(\arcsin(\frac{1}{3}))$$

$$\text{Let } y = \arcsin(\frac{1}{3}) \Leftrightarrow$$

$$\sin y = \frac{1}{3} \quad (y/r) \text{ and } y \in [-\pi/2, \pi/2]$$



$$\therefore \tan y = \frac{1}{2\sqrt{2}}$$

$$\sqrt{3^2 - 1^2} = \sqrt{8} = 2\sqrt{2}$$

Example Find

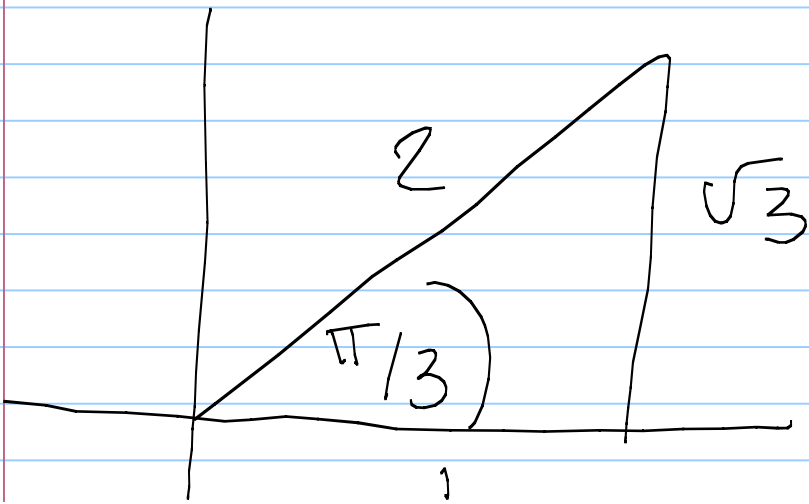
$$1] \sin^{-1}\left(\frac{\sqrt{3}}{2}\right) = \arcsin\left(\frac{\sqrt{3}}{2}\right)$$

$$2] \arccos\left(\frac{1}{2}\right) = \cos^{-1}\left(\frac{1}{2}\right)$$

$$3] \arctan\left(-\frac{1}{\sqrt{3}}\right)$$

$$\textcircled{1} \arcsin\left(\frac{\sqrt{3}}{2}\right) = y$$

$$\Leftrightarrow \sin y = \frac{\sqrt{3}}{2} \quad (y/r) \text{ and } y \in \left[-\pi/2, \pi/2\right]$$



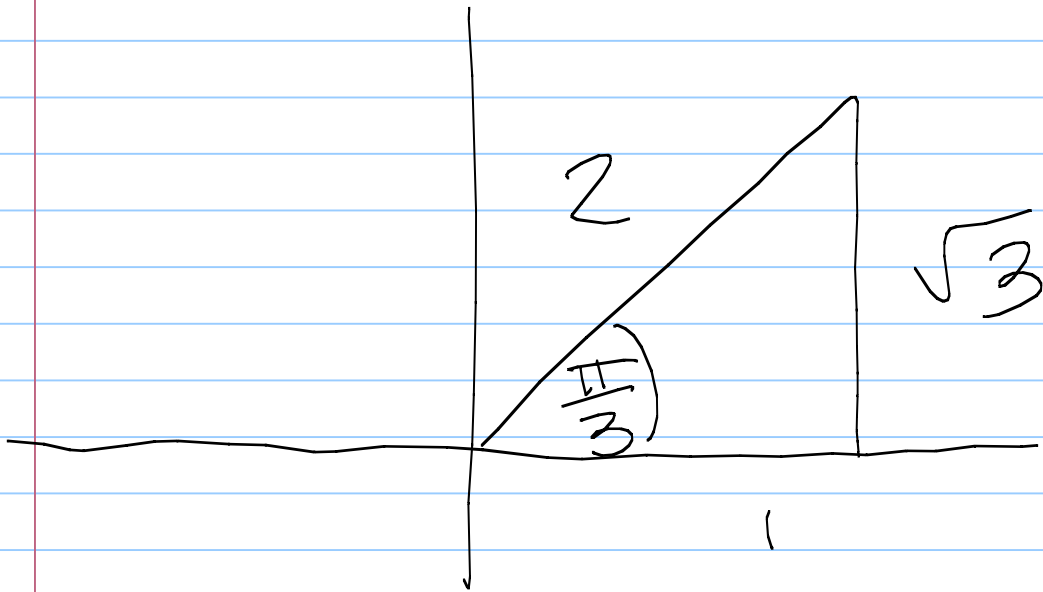
Then

$$y = \pi/3$$

$$\textcircled{2} \quad \arccos\left(\frac{1}{2}\right) = y$$

($\frac{x}{r}$)

$$\Leftrightarrow \cos y = \frac{1}{2} \quad \text{and} \quad y \in [0, \pi]$$

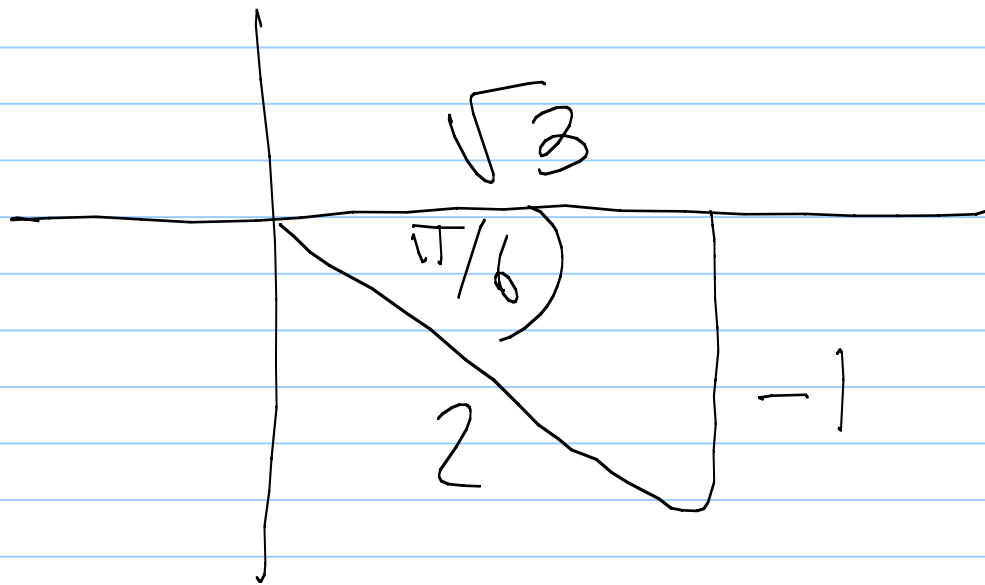


Then

$$\arccos\left(\frac{1}{2}\right) = \frac{\pi}{3}$$

$$\textcircled{3} \quad \arctan\left(-\frac{1}{\sqrt{3}}\right) = y$$

$$\Rightarrow \tan y = -\frac{1}{\sqrt{3}} \quad \text{and} \quad y \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$$



$$y = -\pi/6$$